SKILL WEEK-3

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Aug  8 08:07:58 UTC 2026

  System load:  0.0                Processes:             65
  Usage of /:   0.2% of 1006.85GB  Users logged in:       0
  Memory usage: 6%                 IPv4 address for eth0: 172.21.8.63
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
s
   just raised the bar for easy, resilient and secure K8s cluster deployment
.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/laasya/.hushlogin file.
laasya@laasya:~$ mkdir week3
laasya@laasya:~$ cd week3
laasya@laasya:~/week3$ pwd
/home/laasya/week3
laasya@laasya:~/week3$ nano task1.c
laasya@laasya:~/week3$ ls
task1.c
laasya@laasya:~/week3$ gcc task1.c -o task1
laasya@laasya:~/week3$ ./task1
myshell> hello
You entered: hello
myshell> exit
```

```
This message is shown once a day. To disable it please create the
/home/laasya/.hushlogin file.
laasya@laasya:~$ mkdir week3
laasya@laasya:~$ cd week3
laasya@laasya:~/week3$ pwd
/home/laasya/week3
laasya@laasya:~/week3$ nano task1.c
laasya@laasya:~/week3$ ls
task1.c
laasya@laasya:~/week3$ gcc task1.c -o task1
laasya@laasya:~/week3$ ./task1
myshell> hello
You entered: hello
myshell> exit
You entered: exit
myshell> ^C
laasya@laasya:~/week3$ nano task1.c
laasya@laasya:~/week3$ gcc task1.c -o task1
laasya@laasya:~/week3$ ./task1
myshell> hello
You entered: hello
myshell> exit
Exiting...
laasya@laasya:~/week3$ nano File1.txt
laasya@laasya:~/week3$ ls
File1.txt  task1  task1.c
laasya@laasya:~/week3$ cat File1.txt
This is File1.
This file is created for Task2.
We are copying this content using system calls.
laasya@laasya:~/week3$
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

```
laasya@laasya:~/week3$ nano task1.c
laasya@laasya:~/week3$ gcc task1.c -o task1
laasya@laasya:~/week3$ ./task1
myshell> hello
You entered: hello
myshell> exit
Exiting...
laasya@laasya:~/week3$ nano File1.txt
laasya@laasya:~/week3$ ls
File1.txt  task1  task1.c
laasya@laasya:~/week3$ cat File1.txt
This is File1.
This file is created for Task2.
We are copying this content using system calls.
laasya@laasya:~/week3$ nano task2.c
laasya@laasya:~/week3$ ls
File1.txt  task1  task1.c  task2.c
laasya@laasya:~/week3$ gcc task2.c -o task2
laasya@laasya:~/week3$ ls
File1.txt  task1  task1.c  task2  task2.c
laasya@laasya:~/week3$ ./task2
File copied successfully.
laasya@laasya:~/week3$
```

```
laasya@laasya:~/week3$ nano task2.c
laasya@laasya:~/week3$ gcc task2.c -o task2
laasya@laasya:~/week3$ ./task2
File copied successfully.
laasya@laasya:~/week3$ cat File2.txt
This is File1.
This file is created for Task2.
We are copying this content using system calls.
laasya@laasya:~/week3$
```

Task 3: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

```
laasya@laasya:~/week3$ pwd
/home/laasya/week3
laasya@laasya:~/week3$ nano task3.c
laasya@laasya:~/week3$ ls
File1.txt  File2.txt  task1  task1.c  task2  task2.c  task3.c
laasya@laasya:~/week3$ gcc task3.c -o task3
laasya@laasya:~/week3$ ls
File1.txt  File2.txt  task1  task1.c  task2  task2.c  task3  task3.c
laasya@laasya:~/week3$ ./task3
myshell> hello
You entered: hello
myshell> ./task3
You entered: ./task3
myshell> helllo
You entered: helllo
myshell> hello
You entered: hello
myshell> exit
You entered: exit
Exiting...
laasya@laasya:~/week3$ 
```

Task 4: To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality

```
laasya@laasya:~/week3$ nano task4.c
laasya@laasya:~/week3$ ls
File1.txt  task1    task2    task3    task4.c
File2.txt  task1.c  task2.c  task3.c
laasya@laasya:~/week3$ gcc task4.c -o task4
laasya@laasya:~/week3$ ls
File1.txt  task1    task2    task3    task4
File2.txt  task1.c  task2.c  task3.c  task4.c
laasya@laasya:~/week3$ ./task4
myshell> apple
You entered: apple
myshell> computer
You entered: computer
myshell> hello
You entered: hello
myshell> exit
Exiting...
laasya@laasya:~/week3$ 
```

```
laasya@laasya:~/week3$ nano task5.c
laasya@laasya:~/week3$ ls
File1.txt  task1    task2    task3    task4    task5.c
File2.txt  task1.c  task2.c  task3.c  task4.c
laasya@laasya:~/week3$ gcc task5.c -o task5
laasya@laasya:~/week3$ ls
File1.txt  task1    task2    task3    task4    task5
File2.txt  task1.c  task2.c  task3.c  task4.c  task5.c
laasya@laasya:~/week3$ ./task5
myshell> hello
You entered: hello
myshell> computer
You entered: computer
myshell> hello world
You entered: hello world
myshell> hello
You entered: hello
myshell> This is a very long command used for testing dynamic memory allocat
ion
You entered: This is a very long command used for testing dynamic memory all
ocation
myshell> apple computer university
You entered: apple computer university
myshell> exit
Exiting...
laasya@laasya:~/week3$
```